Prabhavit Gupta

Applied AI Engineer: Autonomous Agents, Operational Workflows & Production Backends



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Summary

Applied AI Software Engineer with 3 years of hands-on experience building, deploying, and operating production AI systems, autonomous agent workflows, and backend data integrations. Proven track record turning ambiguous operational bottlenecks into dependable software: architecting multi-step agents that automate mission-critical customer workflows, grounding models in dynamic operational databases, and owning the complete lifecycle from initial design through automated testing to live observability. Core technical foundation in Python (FastAPI, backend services), TypeScript/Node.js, PostgreSQL, and cloud deployments, with specialized expertise in practical generative AI: schema-enforced tool calling, context engineering, deterministic guardrails, and real-world transcript evaluation. Pragmatic builder operating daily with AI-native workflows (Claude Code, Cursor) who understands when to leverage agents versus deterministic rules to scale operational capacity. Holder of EU Long-Term Residence Permit (permanent unrestricted EU work authorization in Germany without visa sponsorship); based in Venice, Italy and ready for immediate relocation to Berlin.

Technical Expertise

- **Autonomous Agents & Operational Workflows:** Multi-step agent orchestration, tool and function calling, LangGraph, schema-enforced JSON validation, execution sandboxes, guardrails, fallback escalation triggers.
- **System Integration & API Architecture:** RESTful API design, webhook synchronization pipelines, automated contract testing with Playwright, third-party operational service connectors, structured logging.
- **Production LLMOps & Evaluation:** Automated evaluation loops analyzing real operational transcripts, hallucination prevention, latency profiling, token economics, context window management, trace observability.
- **Full-Stack Engineering & Data Systems:** Python (FastAPI, backend services), TypeScript / JavaScript (Node.js, Next.js, React), PostgreSQL, schema design & migrations, Supabase, Redis.
- **Cloud Infrastructure & Ownership:** Docker containerization, AWS/GCP cloud environments, CI/CD pipelines, Git, workflow automation (n8n), end-to-end operational monitoring and incident debugging.
- **Languages & Authorization:** English (C1 Professional), Hindi (Native), Italian (Beginner). EU Long-Term Residence Permit (permanent unrestricted work authorization in Germany and EU, no visa sponsorship required).

Experience

FlowSpace.co **Mar 2024 - Aug 2026**
Lead Developer (Full-time, Freelance) *London, UK (remote)*

- Architected, deployed, and maintained the production AI customer support agent runtime on a live platform serving 4,500+ active users, automating customer inquiries across pricing, venue inventory, and bookings.
- Grounded agent responses against live operational databases and partner APIs (OfficeRnD, Nexudus) using strict JSON tool schemas and fallback escalation logic that routed complex inquiries to staff.
- Built automated evaluation loops analyzing real customer conversation transcripts rather than synthetic benchmarks, systematically detecting failure patterns and eliminating hallucinated answers.
- Profiled latency and API token consumption across model versions, optimizing prompt templates and context window utilization to reduce inference costs while maintaining rapid response times.
- Deployed automated end-to-end regression test suites using Playwright, verifying API contracts and continuous deployment stability across end-to-end customer booking flows.
- Spearheaded platform migration from legacy architecture to a modern Next.js and backend service stack, supporting platform growth from 2,000 to 4,500+ users without operational degradation.
- Built automated data ingestion and transformation pipelines parsing and validating unstructured partner listings, expanding platform inventory to over 900 locations.

Urlink **Apr 2025 - Jul 2025**
Project Manager & AI Assisted Developer (Freelance) *Barcelona, Spain (remote)*

- Architected core web application backend workflows, defining clear API contracts, relational schemas, and data transformation logic for client-facing features.
- Implemented automated validation and synchronization pipelines connecting third-party services with PostgreSQL, preventing malformed inputs and ensuring relational data integrity.

- Integrated automated CI/CD workflows and regression checks to accelerate feature delivery across distributed development environments.

Independent Technical Consultant **Aug 2023 - Feb 2024**
AI Solutions & Automation Engineer *Venice, Italy*

- Designed and delivered bespoke AI automation workflows, web extraction engines, and API integrations for European SME clients using Python, TypeScript, and Playwright.
- Translated manual, fragmented business operations into automated software solutions with structured logging, retry queues, and reliable recovery logic.
- Engineered webhook listeners and real-time synchronization pipelines across CRM platforms, operational tools, and relational datastores.

Education

Ca' Foscari University of Venice **2020 - 2024**
BA in Philosophy, International Studies and Economics *Venice, Italy*

- Rigorous grounding in formal logic, analytical problem solving, systems analysis, and macroeconomic decision models.